

Problem Set 4

Jhinuk Banerji

February 23, 2026

Question 7: Data Sources for Web Scraping

There are some data sources I would be interested in scraping for my research. First, **Federal Election Commission (FEC)** website for observing the fundraising patterns by candidate gender. Second, **OpenSecrets.org** for data on political donations, lobbying expenditures, and candidate profiles. Third, the **Inter-Parliamentary Union (IPU)** for getting country-level data on women's representation in national parliaments over time.

Question 6: sparklyr exercise

Part (b): What is df?

The object df is a Spark DataFrame, specifically a `tbl_spark` object. Unlike a standard R data frame, which is stored in memory, a `tbl_spark` is a distributed dataset stored across Spark's cluster nodes, where computations are executed lazily on the Spark backend rather than immediately in R.

Part (c): Filtered output

After filtering for rows where "Sepal.Length > 5.5", the resulting Spark DataFrame contains only those observations from the iris dataset meeting that condition. The printed output would display a preview of the first few rows and columns.

Part (d): Number of rows

Of the 150 total observations in the "iris" dataset, 71 have "Sepal.Length > 5.5". The `sdf_nrow()` function computes this count by executing the query on the Spark cluster and returning the result to R.